

Chapter 3. Problem Solving by Informed Search

Topic 3.1. AI problems and Heuristic Search Techniques

- a) Chess playing / planning robot navigation / finding language structures / visual perception: searching through a state space for a goal state
- b) Goal-based agents are considered.

Basic Concepts:

1. Initial state

Description of environment that the agent starts with

Classroom / Toy example:
8-puzzle problem

Any placement
of tiles, S_0

6	3	4
7		5
2	1	8

2. Goal state

Description of environment that the agent wants to achieve

An ordered
placement
of tiles, G_i

1	2	3
4	5	6
7	8	

3. Successor function

That takes a state as argument and
returns a set of $\langle \text{action}, \text{state} \rangle$ pairs

$$\text{SUCCESSOR}(S_0) = \{ \langle \text{up}, S_1 \rangle, \langle \text{left}, S_2 \rangle, \dots \}$$

6		4
7	3	5
2	1	8

6	3	4
7	5	
2	1	8

6	3	4
	7	5
2	1	8

6	3	4
7	1	5
2		8

4. State Space

Set of states reachable from the initial state;
Initial state + successor function

$S_0, S_1, S_2, \dots$
 $S_0, \text{SUCCESSOR}$

5. Path

A sequence of states connected by actions

$S_0 - \text{up} - S_1 - \text{right} - S_{12} \dots$

6. Goal test

Test to determine whether a given state is a
goal state or not

Matching with a given
possible goal state

7. Path cost

Sum of step costs

Uniform / non-uniform

Number of steps / actions in the path; Uniform step cost, 1;

$$c(n, a, n') = c(S_0, \text{up}, S_1) = 1$$

8. Solution

A path from the initial state to a goal state

A path in a directed graph or tree constructed by SUCCESSOR, from S_0 to G_i .

9. Optimal Solution

Solution that has the lowest path cost

A specific path, P_s found in the directed graph.

c) Conventional vs Heuristic search

- Search space may be very big
- Efficient algorithms needed
- Uninformed or blind search not usable directly
- Informed or Heuristic search helps
 - Additional prior knowledge
 - Estimation or problem specific rule
 - Helps sometimes, but not always
 - Heuristics: 'to find', 'to discover', 'rule of thumb' or 'judgmental technique'

d) Heuristic Functions

Heuristics is often expressed as a function of a state:

$h(s)$ = Estimated cost of the cheapest path from s to a goal state.

Examples:

i) 8-puzzle problem

Select the generated state s with

$h_1(s)$ = number of tiles not matching those in the goal.

6	3	4
7		5
2	1	8

Initial

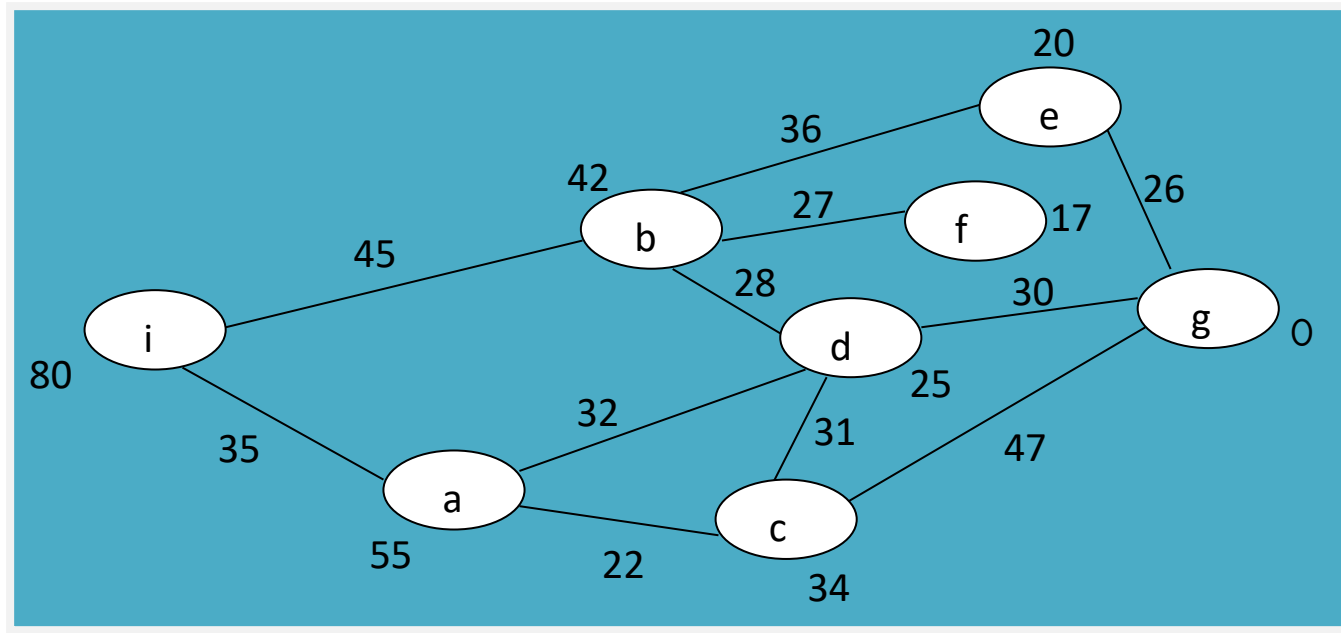
	1	2
3	4	5
6	7	8

Goal

$$h_1(\text{Initial}) = 6$$

ii) For finding the shortest path between two towns intended for traveling through other towns:

$h_2(s) = \underline{\text{Straight-line distance}}$ from town s to the destination.



i: Initial state/node
(source)

g: Goal state/node
(destination)

$h_2(i) = 80$

e) Good heuristic functions

- Finding a good heuristic function is a big problem.
- Statistical and probabilistic methods help.
- May require additional computational cost (as for h_1), or prior knowledge (as for h_2).
- Good heuristic functions are said to be admissible or consistent.

f) Admissibility of heuristic functions

☐ An admissible heuristic function never over estimates the cost to reach the goal:

$$h(s) \leq \text{Actual cost to reach the goal from } s$$

☐ h_1 and h_2 seen above are admissible:

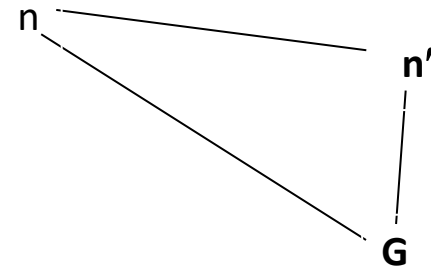
h_1 – any tile that is out of place, must be moved at least once

h_2 – straight line is the shortest of any other lines

g) Monotonicity or Consistency of heuristic functions

$h(n) \leq c(n, a, n') + h(n')$, for any node n and every successor n' of n

It is from triangular inequality:
one side is shorter than the
sum of the other two.

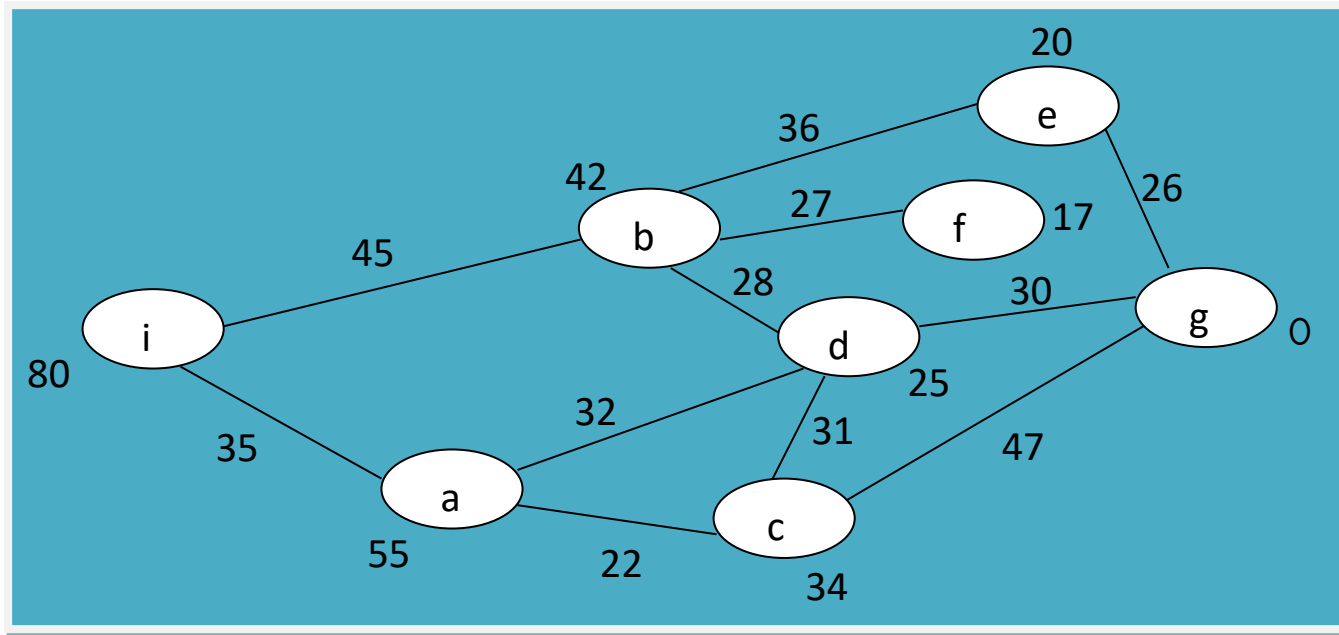


❖ h_1 and h_2 seen above are also consistent.

❖ Every consistent heuristic function is also admissible, but vice-versa is not always true.

[May work very well on n' , while not so well on n .]

h) Checking for admissibility and consistency of heuristic functions



$h(s) \leq \text{Actual cost to reach the goal from } s, \text{ for any node } s$

$h(i)=80 \leq \text{cost}(i-b-e-g)=107 \mid \text{cost}(i-b-d-g)=103 \mid \dots;$

...

$h(n) \leq c(n, a, n') + h(n'), \text{ for any node } n \text{ and every successor } n' \text{ of } n$

$h(i)=80 \leq c(i, \rightarrow, b) + h(b)=45+42, h(i)=80 \leq c(i, \rightarrow, a) + h(a)=35+55;$

$h(b)= \dots; \dots$